IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for Charge Sensitive Preamplifiers with Feedback Resistor

PROCEDURE Nº MRNI-509
REV. D0

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1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the procedures employed in the test and measurement of the characteristics of charge sensitive preamplifiers with feedback resistor.

1.2.- Scope

This description is applicable to charge sensitive preamplifiers with feedback resistor, commonly used with radiation detectors in nuclear applications.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

NIM	Nuclear Instrumentation Modules.
DMM	Digital Multimeter
R	Energy Resolution
FWHM	Full width at half of maximum
CSP	Charge sensitive preamplifier
INL	Integral non linearity.
rms	Root mean square value
ENC	Equivalent noise charge

2.2.- Definitions

2.2.1. Detector
A device that converts the energy of a photon or incident particle in an electric pulse.

2.2.2. Noise
Any unwanted signal that masks the signal of interest in detection systems.

2.2.3 ENC
Noise in units of rms charge for a preamplifier, reflected to the input.

3.- INTRODUCTION

Charge sensitive preamplifiers, CSP, are commonly used to handle the nuclear pulses generated in radiation detectors. This pulses are very small and therefore the CSP needs to have a very high "gain" and low noise.

For example, in a cooled (77 °K) Germanium detector, the energy required to produce an electron-hole pair is 2.96 eV, then for a photon of 661.66 KeV from Cs-137, the number of produced pairs is 223 533.78, which have a charge ($\times$ 1.6 x10^{-19} C) of 3.58 $\times$ 10^{-14} C.

The voltage amplitude of the output pulse Vo, see Fig. 1, in a CSP is proportional to the input charge, Qi, generated in the detector, and does not depend on the detector capacitance.

$$V_o = - \frac{Q_i}{C_f} \qquad (1)$$

In the example, if the CSP has a capacitor with a value C_f = 1 pF, the output voltage for the Cs-137 pulse will be 35 mV.

In the basic circuit of a CSP the charge will be accumulated in the feedback capacitor Cf , until the voltage across the capacitor Vc (Vc ≈ Vo), reaches the saturation value according with equation (1).

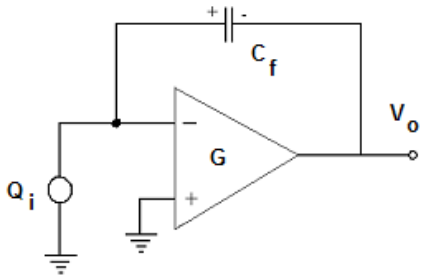


Fig. 1. Basic circuit of a CSP.

In order to continue processing the input pulses, C_f needs to be discharged, there are two basic modes in which the capacitor can be discharged: the dynamic discharge with a resistive feedback, see Fig.2, and the reset discharge with the application of a fast pulse.

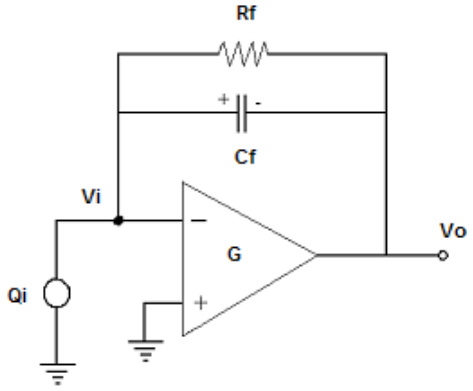


Fig. 2. Circuit of a CSP with resistive feedback.

In the resistive feedback, the soft discharge, with a decay constant given by the product of Rf and Cf, occurs for every input pulse, see Fig. 3.

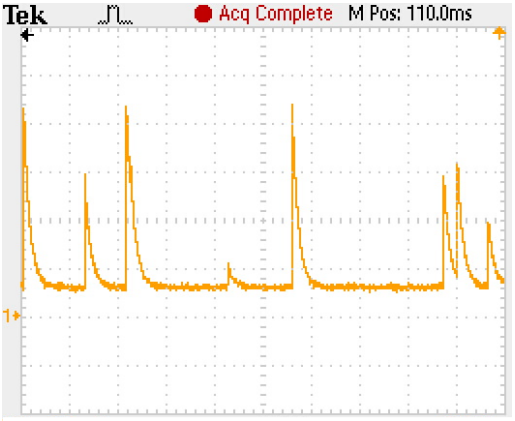


Fig. 3. Output signals of a CSP with resistive feedback.

In a commercial CSP, an output amplifying stage with a voltage gain A1 is added after the main stage, see Fig. 4. Also an internal pole–zero cancellation circuit, Rk-C1, is included.

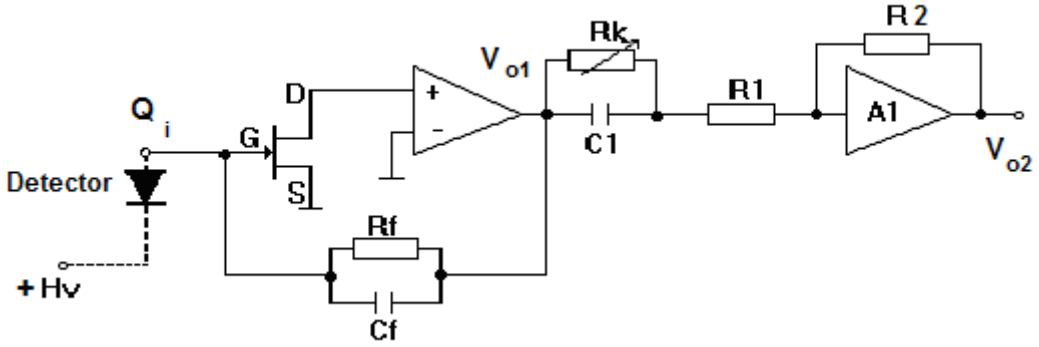


Fig. 4. A complete CSP with resistive feedback.

therefore the total conversion gain is:

$$G_{CT} = -\left(\frac{V_{o1}}{Q_i}\right)\left(\frac{V_{o2}}{V_{o1}}\right) \qquad (2)$$

$$G_{CT} = -\left(\frac{V_{o1}}{Q_i}\right) A_1 \qquad (3)$$

4.- DEVELOPMENT

The sequence of steps to verify the electrical characteristics and operation of CSP with feedback resistor is described in the next paragraphs. A flux diagram of the process is shown in Annex I.

4.1- Generalities

Whenever possible, refer to the test conditions recommended by the manufacturer in the sheet of specifications, a technical report about the verification of the electrical characteristics of the CSP must be elaborated, including the circuit diagram, environmental conditions, testing set-up, and details of the test instruments.

4.1.1.- Verify that the CSP is installed and connected according to the instructions of the manufacturer.

4.2.- Test Instruments

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

4.2.1 Pulse generator.

Produces voltage tail pulses with a precise relative amplitude that is selected with a calibrated dial and attenuation steps.

4.2.2 Spectroscopy amplifier.

Amplifies tail pulses and includes a semi-gaussian shaping in the pulse, use the unipolar output.

4.2.3 Oscilloscope

Use an analog or digital oscilloscope.

4.2.4 MCA

Use an MCA at least with 1024 channels

4.3- Test Conditions

Generally the CSP needs to reach its stable temperature, in order to have its best performance, a warming period of half an hour is enough.

4.3.1.- Temperature

A stable temperature is recommended to perform the tests, between 20º C and 25ºC.

4.3.2.- Humidity

The humidity must be less that 50 % of relative humidity, because the preamplifiers are very sensitive and the accumulation of humidity in the input could modify the isolation conditions.

4.4.- Measurements

4.4.1. Characteristics of the output pulse.

Assemble the set-up of Fig. 5, apply an input pulse of 200 mV, adjusting in the generator its rise time to the minimum (50 ns or less) and the fall time to 50 μs. Measure the rise

time *tr*, see Fig. 6, and the fall time *tf* see Fig. 7, at the output of the preamplifier with the oscilloscope.

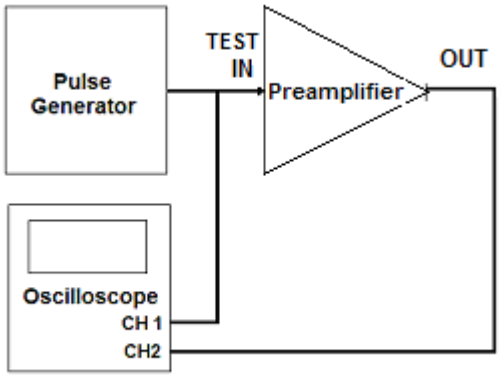


Fig. 5. Set-up for the measurement of the CSP output pulse.

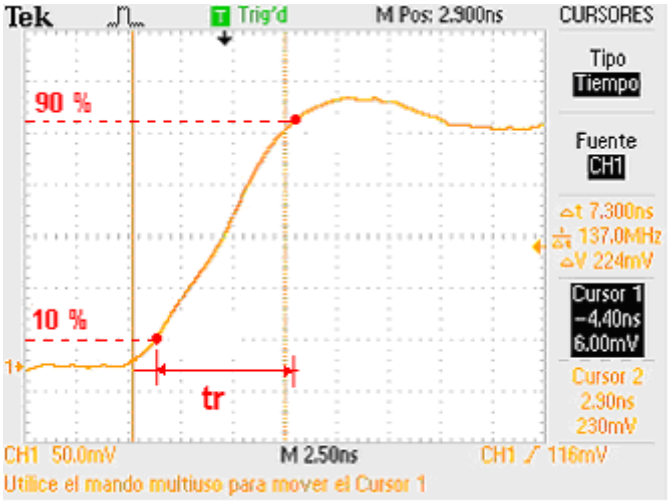


Fig. 6. Measurement of the rise time, tr, of the output pulse.

The rise time indicates how fast the preamplifier is, it must be as small as some ns. It is measured from 10% to 90% of the amplitude of the pulse, with the obtained value, the bandwidth BW, of the CSP can be evaluated:

$$BW \cong \frac{0.35}{tr} \quad [\text{ Hz }] \qquad (3)$$

In the example, *tr* = 6.2 ns, and the BW of the CSP is approximately 56 MHz.
The fall time depends of the design of the preamplifier but it is around some μs, in this case, *tf* = 90 μs.

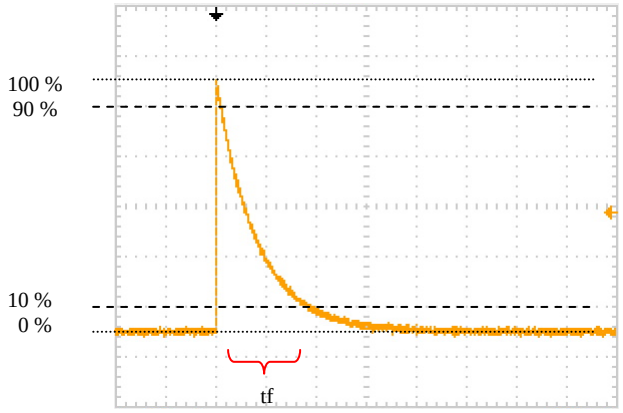


Fig. 7. Measurement of the fall time, tf, of the output pulse, the settings of the oscilloscope are: 50 mV/div, 50 μs/div.

4.4.2. Measurement of the conversion gain G_C.

The conversion gain G_C is defined as:

$$G_C = \frac{Vo}{Qi} = -\frac{1}{C_f} \qquad \left[\frac{V}{C}\right] \tag{4}$$

- Assemble the set-up of Fig. 5, apply an input pulse V_i = 200 mV.
- If the test capacitor inside the preamplifier is C_t, the input charge applied Q_i, is:

$$Q_i = C_t\, V_i \tag{5}$$

- Measure the output voltage V_o, and calculate G_C with the equation 4.
  For example, if C_t is 1.5 pF and if the measured value Vo is 300 mV, then:

$$G_C = 1\,\frac{V}{pC}$$

- The conversion gain also can be specified in energy values applying the equation:

$$G_E = -\frac{e\,G_C}{\varepsilon} \tag{6}$$

where: G_E is in [V/eV].

ε_G= 2.97 eV / Par (for a Germanium detector)

ε_S= 3.96 eV/ Par (for a Silicon detector)

e = 1.6×10^{-19} C

calculating with the values of the former example for a germanium detector:

$$G_E = -0.035\,\frac{V}{MeV} = -35\,\frac{mV}{MeV}$$

4.4.3 Integral non linearity, INL

The INL is the maximum deviation from an ideal linear relationship between an output and an input in a system considered linear. The measurement of the INL is performed considering as reference the good precision of dial in the generator, follow the next steps:

- Use the experimental set-up of Fig. 5.
- Fix the main dial of the generator in 9.0.
- Adjust the amplitude of the precision generator to get a pulse with the maximum voltage.
- Measure the output pulse of the CSP and register the value in the Table 1 of the Test Report.
- Select at least 16 equally spaced values down to 1.0 in the dial, measure and register the amplitude of the output pulse of the CSP for every value of the dial.
- Analyze all the values of Table 1 with a reliable software, for example ORIGIN, to get the linear fit and the deviation of voltage values with respect to the ideal ones obtained from the linear fit. See Fig. 8.

$$Y = m\,x + b \qquad (7)$$

Where:

Y = amplitude values
m = slope of the line, V/ dial values
x = value of the dial
b = theoretical amplitude for dial=0

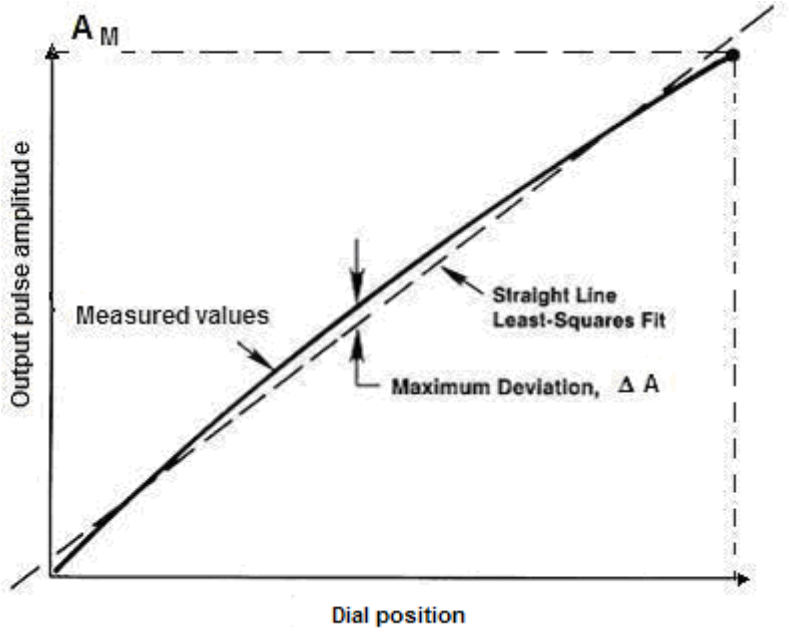


Fig. 8. Linearity of a CSP referred to the dial of the precision generator.

- For every dial value we have the ideal value of the output voltage V_d and the corresponding measured value V_r. The maximum deviation, ΔV_{max}, obtained along the total range V_{max}, defines the value of the INL in per cent:

$$INL = \frac{(V_d - V_r)\max}{V_{\max}} \times 100 \qquad (8)$$

Where:
V_r – measured value of the output voltage.
V_d – ideal value of the output voltage obtained from the linear fit.
V_{max} – maximum output voltage of the CSP.
$\Delta V_{\max} = (V_d - V_r)\max$

The former method can give an approximated value of the INL, in order to get a more accurate value, a spectroscopy amplifier and a MCA can be used, assuring that the INL of these instruments is within its specifications as stated by the manufactures in the technical manual. In this case follow the method described in the procedure: MRNI-514.

4.4.4 Noise.

The noise of the electronic components of a radiation measuring system becomes important for spectroscopy applications because it can spoil the energy resolution of the system. The total resolution FWHM in eV, is related with their constituents as:

$$FWHM^2 = RD^2 + RS^2 + RE^2 \qquad (9)$$

Where:
RD –is the contribution to resolution of the noise of the detector, includes mainly the effect of the leakage current, in a detector with good conditions this component could be neglected.
RS - is the contribution to resolution of the statistical distribution of the energy values in the detector.

$$RS = 2.35\sqrt{\varepsilon \quad F \quad E} \qquad (10)$$

Where: F = Fano Factor ≅ 0.1
E = Energy of the radiation in eV

RE - is the contribution to resolution of the associated electronics and is generally dominated by the noise of the input stage, the preamplifier, and the detector capacitance.

a) Noise measurement with the complete system.
The noise of the complete system can be obtained at the output of a spectroscopy chain. See Fig. 9. The settings of the system are the same as the used in the specific application.

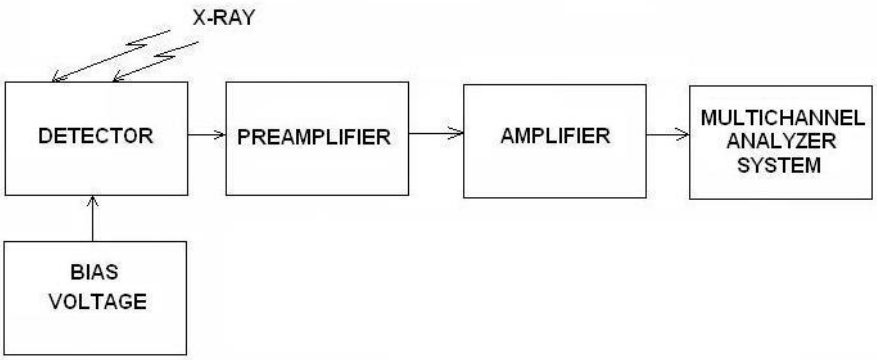


Fig. 9.- Block diagram of a spectroscopy system

The next steps can be followed:

- Calibrate the system in energy (refer to the applicable procedure, MRNI-505, MRNI-506, MRNI-507 or MRNI-508).
- Accumulate approximately 10000 counts in the reference peak (Si: 5.89 keV or 5.489 MeV, Ge: 1332.5 keV) and obtain the FWHM.
- Calculate RS with equation 10.
- Neglecting the value of RD, the value of RE is:

$$RE = \sqrt{FWHM^2 - RD^2} \qquad (11)$$

- The noise of the preamplifier referred to the input in equivalent charge units, ENC, in rms electrons (root mean square value), is calculated as:

$$\mathrm{ENC} = \frac{\mathrm{RE}}{2.35\ \varepsilon} \qquad (12)$$

For example, in a Ge detector, the calculated RS = 1.478 keV for an energy of 1332.5 keV and if the measured resolution was 1.8 keV, then RE = 1.027 keV and finally, ENC = 147.2 electrons (rms)

b) Noise Measurement of the Preamplifier.

When the detector can be separated from the preamplifier, the next steps can be followed:

- Measure the value of the G_C, as in paragraph 4.4.2.
- Use the set-up of the Fig. 10 and apply a pulse of 50 mV in the test input of the preamplifier. The settings of the amplifier are the same as the used in the specific application but the gain must be enough to have a pulse of around 5 V.
- Measure the pulses at the output of the preamplifier, Vp1, and amplifier Vp2. To calculate the gain G.

$$\mathrm{G} = \frac{\mathrm{Vp2}}{Vp1} \qquad (13)$$

- Turn off the generator and measure the peak to peak value of the output noise voltage Vno, at the output of the amplifier, with the oscilloscope.

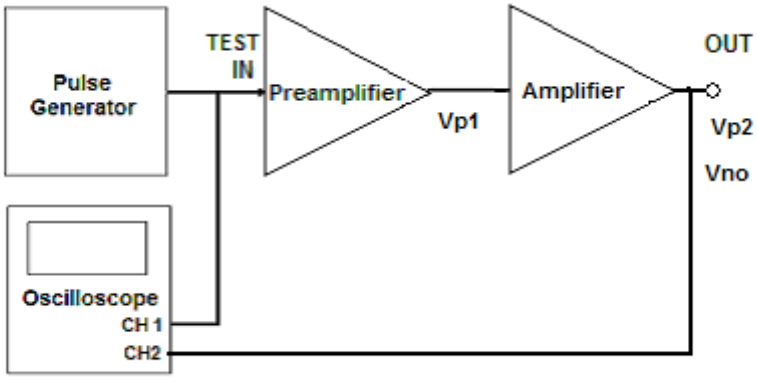


Fig. 10. Set-up for the Noise Measurement of the Preamplifier.

- Considering that the measured signal is white noise, calculate the output noise in rms value, Vno(rms):

$$\text{Vno(rms)} = \frac{\text{Vno}}{6} \qquad (14)$$

- Calculate the ENC in electrons rms, with the equation 15:

$$\text{ENC} = \frac{\text{Vno(rms)}}{G_C \quad G} \qquad (15)$$

5.- ADMINISTRATION OF THE TEST REPORTS

5.1 Numbering of the reports.

All the generated technical test reports must have a unique and consecutive number.

5.2.- Personnel.

The test of detectors and electronic modules must be done by trained personnel.

5.3.- Technical Report of results.

5.3.1 The results of the test of detectors and electronic modules must be registered in a unique test report, stating the description of the detectors or/and electronic modules, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

5.3.2 All the test reports must be classified and keep in a folder for future consult.

6.- ACTION IN CASE OF NON CONFORMITIES.

6.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

6.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

7.- RESPONSIBILITIES

7.1.- Head of the Department.

Supervise that all the activities for testing of detectors and electronic modules follow the established procedure.

7.2.- Area Responsible.

7.2.1 Assure that all the electronic test equipment be in good operational conditions and calibrated.

7.3.2 Verify that all the activities for testing of detectors and electronic modules follow the established procedure

7.2.3 Verify that the test reports contain all the details of the testing of detectors and electronic modules.

7.2.4 Maintain a register and control of the test reports for all the detectors and electronic modules tested in the laboratory.

7.**3 Operative Personnel.**

7.3.1 Verify that all the electronic test equipment be in good operational conditions and calibrated.

7.3.2 Follow the steps established in this procedure for the testing of detectors and electronic modules.

7.3.3 Elaborate the test report of all the tests of the detectors and electronic modules.

7.3.4 Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

8.- BIBLIOGRAPHY

1.- IAEA "Selected Topics on Nuclear Electronics" IAEA TECDOC 363, Vienna, 1987.

2.- IAEA "Nuclear Electronics Laboratory Manual" IAEA TECDOC 530, Vienna, 1987.

3.- Princeton Gamma Tech, Inc., "Detector Installation and Optimization" USA, 1980.

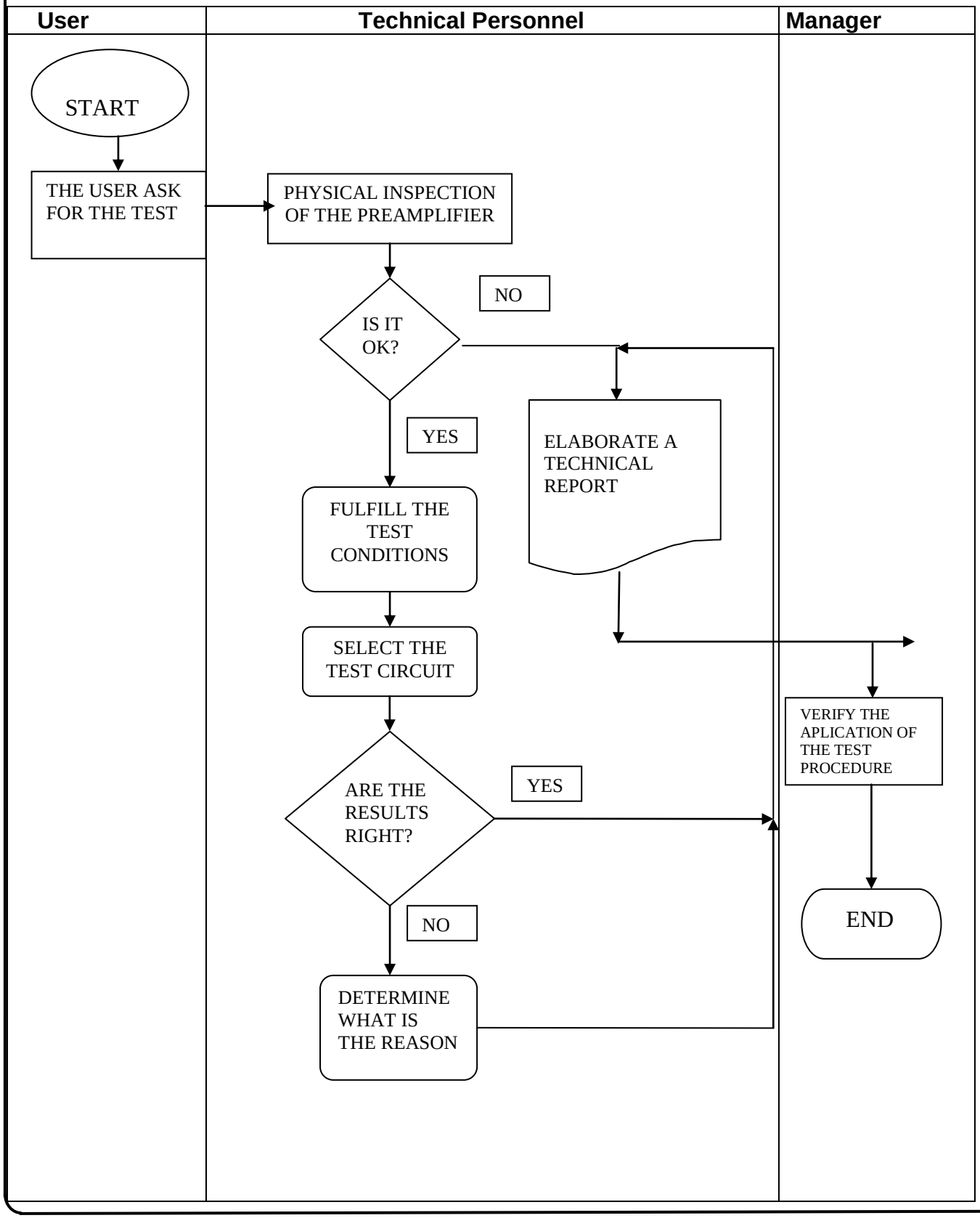
4.- IEEE "Standard Test Procedures for Amplifiers and Preamplifiers used with Detectors of Ionizing Radiation", IEEE Std 301-1988, (Revision of ANSI/IEEE Std 301-1976), 1988.

9.- ANNEXES

Annex I . Flow Chart
Annex II. Test Report

Annex I

Flow Chart.


User
Technical Personnel
Manager
START
THE USER ASK FOR THE TEST
PHYSICAL INSPECTION OF THE PREAMPLIFIER
IS IT OK?
NO
YES
ELABORATE A TECHNICAL REPORT
FULFILL THE TEST CONDITIONS
SELECT THE TEST CIRCUIT
ARE THE RESULTS RIGHT?
YES
NO
DETERMINE WHAT IS THE REASON
VERIFY THE APLICATION OF THE TEST PROCEDURE
END

Annex II

Test report

TEST REPORT Nº PFR-_____

Preamplifier number:_____

Preamplifier Mark:_____ Model:_____
Serial number:_____For:_____detector

Physical Revision

Condition	YES	NO
Damage in the Body		
Integrity of the connectors and cables		

Instruments Employed

Instrument	Mark	Model	Serial number
Detector			
Voltage Power Supply			
Pulse generator			
Oscilloscope			
Amplifier			
MCA			

Environmental Conditions

Temperature	
Humidity	

Output Pulse

Rise time, tr	
Fall time, tf	
BW	

Conversion Gain

G_C		
	Vo =	
	Input charge, Qi =	
G_E	Test capacitor, Ct =	

Integral non linearity, INL:

Dial setting	Measured Voltage (Vr)	Ideal Voltage (Vd)	ΔV = Vd - Vr

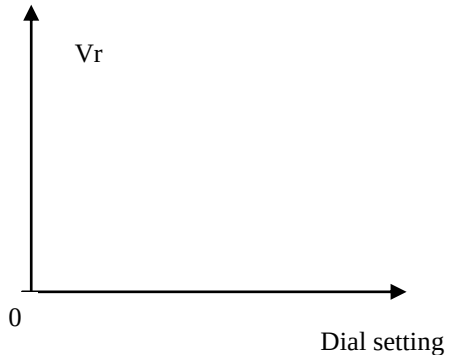


Table 1. Values to calculate the INL.

Noise

Time constant of the amplifier	
Energy, E	
FWHM	
RS (eV)	
RE (eV)	
Detector material	
ENC	
Conversion gain, G_C	
Amplifier gain, G	
Output noise, Vno	
Vno(rms)	
ENC	

Diagnostic or Comments:

Tested By:	
Date:	